

## 1. Introduction

Incident scenarios are commonplace every day. Nature of the scenario, scale, and severity are a few factors that determine the type of response needed to safely address it, without severe consequences. Over the years there seems to have been an increase in active shootings across the country of United States (Katsiyannis et al., 2022). These high stress events cause mass hysteria and panic, as well as several long-term effects in the victims. To respond to these events, various active shooter preparedness training and first responder communities have been enforced and established. These aim to provide necessary measures to try and handle such situations and aim to eliminate the casualty numbers.

One of the most popular methods to mitigate incident scenarios is educating the public about the “Run-Hide-Fight” method. (U.S. Department of Homeland Security, n.d.-a.) The basic idea behind this method is that when faced with an imminent threat (usually an active shooter), the involved people (targets) have three choices in the moment – choosing to run, would mean to actively create space and move away from the threat, choosing to hide, would mean to move away from or obscure oneself from the line of sight of the active threat, and finally choosing to fight, would mean to tackle the threat head on, in hopes to overcome it. These mechanisms are meant to be followed by the individual based on their current evaluation, thus meaning that there is no “one-size-fits-all” answer to such incidents.

Humans are also complex beings, with their lifetime of experience, that shapes their decision making and mentality, at every stage of their life. Personality has been found to have a key involvement in decision-making processes. The behavior of a person has also been found to be influenced by the personality traits they display. Thus, in such cases, it is important to understand that when trying to determine outcomes of choice-based scenarios, the personality traits could be having significant impact on the individual.

## 2. Literature Review

### 2.1. Incident Response Methods – Run Hide Fight

The past few years have seen a concerning increase in the amount of mass shooting incidents in the United States. In the year 2023 itself, 241 incidents have been recorded with at least a singular loss of life and multiple injured personnel (Gun Violence Archive, 2023). Since the 1999 Columbine school shooting (History.com Editors, 2009) there have been 386 school shootings till June 2023 (Cox et al., 2018) which have resulted in major loss of life. In 2021, a bill was introduced in Congress titled “H.R.5248 - School Shooting Safety and Preparedness Act” (*School Shooting Safety and Preparedness Act 2021*) which requires the Department of Education to publish an annual report on the crime and safety indicators, that covers information such as number of nationwide school shootings, motivation of shooter, casualties, threat termination, etc. and has called for the evaluation of measures during such incidents to develop a communication and response plans with the proper agencies. This seems to be a step in the right direction, and more calls to action as such are helpful, however progress seems to be slow in implementing such measures since the last action on the bill was in 2022.

Melmer et. al. (2019) (Melmer et al., 2019) in their study about “mass casualty events (MCE)” studied the role of interdisciplinary response bodies’ emergency preparedness approach during such unpredictable

events. The research focuses on worldwide MCE and provides a high-level approach to emergency response and triage setup methodologies. Measures such as establishment of early partial lockdown, coordination of trauma response and care of critically injured (via the trauma medical director), supervision and documentation of response teams, as well as conducting simulation exercises, could prove pivotal in proper and efficient response to unpredictable events.

The Cybersecurity & Infrastructure Security Agency (CISA) has implemented various active shooter preparedness techniques, which can help first responders and on-scene individuals understand the basics to survival in such intense and unpredictable scenarios. (U.S. Department of Homeland Security, n.d.-a.) The site details Active Shooter Emergency Action Plan, which lays out multiple steps for preparedness for such events (including steps such as forming a team, developing a plan, conducting risk assessments, establishing goals, assessing courses of action, drafting and evaluation, and training of personnel) as well as a “How to Respond” guide (Hughes et al., 2012). This is where there is the first mention of the Run-Hide-Fight technique.

The Run-Hide-Fight (RHF) technique was developed by the city of Houston, Texas, in efforts to lower the casualties suffered during a mass crisis scenario. (U.S. Department of Homeland Security, n.d.-b), In this methodology, the unarmed individuals facing an imminent threat are given 3 choices: detect the nearest exits and “run” away from the shooter, “hide” from the shooter at a safe spot, or “fight” the shooter as a last resort. However, even though this concept has been adopted by numerous agencies (U.S. Department of Homeland Security, n.d.-c.) there is a lack of substantial evidence that can confirm the effectiveness of RHF. Academic literature on effectiveness RHF is also low, as much of the data is anecdotal and mostly is community reports and instruction based – mostly due to the circumstances required for the true application of RHF, which could impact mental health of the participants (Briggs & Kennedy, 2016). Concepts such as Just-In-Time-Training (Monden, 1998) and a study conducted by Farmer & Zhang (2020) point towards the fact that face-to-face instruction on RHF and in-place training helps in making individuals understand the concepts and its importance, however there are severe gaps when it comes to verifying the effectiveness, retention, and execution of such concepts during an actual MCE.

## 2.2. Simulation and Agent Based Modelling of Crisis Response

Simulation is one of those areas that can help in modelling situations that have high physical implementation costs, but the results could provide deeper insights into the existing structures (Crooks & Heppenstall, 2011). Sloan (2011) in their research stated that one of the biggest challenge of disaster response is trying to predict and research such Multi Casualty Incidents (MCIs) due to the unpredictable nature. They stated that a lack of training and education was the primary reason for failure in handling MCIs. Such pitfalls can be detected and addressed if there was a more pro-active approach to understanding such events without incurring the major human, mental, and physical costs associated – something that simulations and agent-based modelling offers.

Melmer et. al. (2019) (Melmer et al., 2019) in their study state that simulation exercise and drill could play a vital role in performance improvements in medical service providers during a mass casualty event. They state that it improves knowledge during a potential MCE. A major benefit of simulation is also its ability to detect potential failures and gaps in implementation, as well as limitation detection, so that improvements and countermeasures could be set up in place. However, a major drawback pointed out is the long-term retention of skills learnt during the simulation training.

In similar fashion, Jacobs & Burns (2017) in their survey of healthcare and public professionals about active shooter events in hospitals, stated that the responses pointed towards the perception of high preparedness of hospitals in an active shooter event, which classified them as having the lowest risk level when compared to shopping center, airport, and public school. However, it was noted that most of the respondents have an expectation that the healthcare professionals should protect the patients, however a third of the respondents do agree that such actions should be a voluntary personal choice and not a requirement. In addition, most respondents believe that repetitive practice as well as simulated active shooter drills enable healthcare professionals to better deal with such scenarios. This shows that even if the perceived risk of an MCE is low, simulations are generally considered to have an overall positive impact in crisis response.

One of the simulation methods that call for the need of human beings conducting certain processes is agent-based modelling (ABM). (Crooks & Heppenstall, 2011; Jackson et al., 2017) In such simulation method, the goal is to mimic a real-world system, in efforts to study its behavior – both with the internal components as well as exposure to any external components. The core of such simulation lies not in direct simulation of an agent, but rather as an emergent macro-level behavior of the entire population, due to micro-level processes (Hughes et al., 2012). Jackson et. al. (2017) in their research on using agent-based modeling for social psychologists (Jackson et al., 2017) state that ABM is well suited to draw insight for questions where individual behavior interactions among each other produce a collective phenomenon. Thus, in a situation such as an active shooter scenario, where there are a host of individuals interacting with each other and the environment around them, in an intense situation, ABM could prove to deliver pivotal insights into understanding group dynamics and then addressing any potential gaps or needs that may rise. This is also a safer alternative to exposing humans to real-life stressful scenarios, in the name of research. Such real-world scenarios have potential for mental anguish in the participant, as found out by Fernandez-Pacheco et. al. (2018).

Nieto Fernández-Pacheco et al. (2018) in their study of analyzing performance of participants when exposed to a simulation of an MCI found that participants faced increased levels of stress – the impact was found to be more significant in participants who performed triage duties. The key highlight was that even though the group with a higher education (Masters) performed well during the study, there was no significant increase in level of stress – thus nullifying the factor of education when faced with an MCI scenario. Furthermore, it was found that since participants performing triage, due to the need for making unfamiliar decisions in that role. No correlation was found between the ability to implement the training lessons learnt prior to the simulation and the levels of stress displayed. One limitation stated was that the simulation was lacking variations (non-perceived) which could be tackled by the method proposed by this paper with agents based on a personality trait. This could be an indication that there is a need to model the personality of the responder (agent or real human) which could play a vital role in an MCE, such as school shootings.

Lee (2019) in their thesis provided valuable insights into agent-based modeling's effectiveness in assessing the Run-Hide-Fight tactic stated by CISA (U.S. Department of Homeland Security, n.d.-c). This research provides insight into implementing an ABM technique in the Columbine school shooting, to understand whether simulation could provide an alternative response by the unarmed individuals, to lower the casualty rate. It would be interesting if this research was augmented by having the agents display a personality to flesh out the logic based on a digital personality. In their research, it was found that the most successful scenario was the "All Fight" scenario where all the students and staff overpowered the

nearest shooter (resulting in a 35.7% higher survival probability than the original incident) (Lee, 2019). Only offensive measures put in place were able to reduce the casualty count from the original incident.

## 2.3. Personality Models

The human personality is a complex and dynamic relationship between various traits that an individual displays. A lot of these are cultivated over time, and are shaped by life experiences, physiological conditions, mental conditions, and various other factors (Klimstra, 2012; Kashima, 2007). Some experiences are more influential than others, and thus make an individual more oriented towards a certain set of characteristics and traits. This is important to understand as the various decision-making processes that occur in one's lifetime are a product of this combination and permutation of traits and characteristics (Mendes et al., 2019; Ozer & Benet-Martínez, 2006). The major models that are being considered for developing the personal traits of agents are: The Big 5 Personality Model (McCrae & Costa, 1987; McCrae & John, 1992), Maslow's Hierarchy of Needs (McLeod, 2007; Maslow, 1943), and DiSC® Model (Sugerman, 2009; Slowikowski, 2005).

The Big 5 Model (commonly referred to as the OCEAN model) includes 5 components according to (Digman, 1990):

1. Openness: "Degree of intellectual curiosity, creativity, and preference for novelty"
2. Conscientiousness: "Tendency to be organized and dependable"
3. Extraversion: "Tendency to seek the company of others and talk"
4. Agreeableness: "Measure of one's trusting and helpful nature"
5. Neuroticism: "Predisposition to psychological stress"

Maslow's Hierarchy of Needs categorizes the needs that humans have, into a pyramid shaped broader categories (namely 5). The base of the pyramid states the absolute necessities for life to sustain properly and the top of the pyramid focuses on the abstract self needs, needed for one to create their identity in society. These needs are listed as follows (Pandya, 2014):

1. Physiological – Essential for humans to sustain life properly and are considered as the base necessities. E.g. – Food, water, sleep etc.
2. Safety – Essential for humans to comfortably live in a safe environment
3. Love/Belonging – Essential for humans to build connections and involves feelings of belongingness, as well as create relationships with other humans
4. Esteem – Essential for a human to place themselves in the society with other humans, and uphold themselves to a standard of living
5. Self-Actualization – Essential for humans to develop a personality and decide how they will shape their perceived view by others, when faced with challenges and decision making (simple to complex, mundane to exciting)

Finally, the DiSC model is responsible for classifying the four main behavioral styles of humans. These can be classified as:

1. Dominance (D): The human emphasizes accomplishment of results. Typical traits include confidence, bluntness, and being demanding.
2. Influence (i): The human emphasizes persuasion of others. Typical traits include enthusiasm, optimism, and trustability.

3. Steadiness (S): The human emphasizes cooperation among everyone and dependability on others. Typical traits include calmness and patience.
4. Conscientiousness (C): The human emphasizes quality and accuracy of results as well as competency. Typical traits include independence and detail oriented.

Along with these four major categories, there also exists the four overlapping categories that distinguish people who display a little of two categories (Di, iS, SC, CD). Using this method, the morality and decision making of an individual can be explored, as well as their interaction with other humans.

Even though there is only one model being used in the proof of concept, it is important to realize the full trait inventory that an agent possesses to have proper interaction among all the variables, as well as infer the correct values, when it comes to “decision making” by the agent.

### 3. Methodology

#### 3.1. Trait Inventory

For this proof of concept, only one personality trait will be considered from the Big 5 Personality Model, and the framework will be developed around that.

Based on the model, the closest personality trait during an intense decision-making scenario was determined to be “Neuroticism” since previous studies have linked poor decision making with elevated levels of this trait (Lauriola & Levin, 2001; Denburg et al., 2009; Byrne et al., 2015). Neuroticism includes various characteristics, which contribute towards determining the strength of the trait in an individual’s personality. For such purposes, a trait inventory was developed for each agent with the following characteristics:

| Variable name | resilient/confident           | sensitive/nervous                   |
|---------------|-------------------------------|-------------------------------------|
| adp           | Deals well with stress        | Experiences a lot of stress         |
| wrd           | Doesn’t worry much            | Worries about many different things |
| sdv           | Rarely feels sad or depressed | Gets upset easily                   |
| wrd           | Emotionally stable            | Experiences dramatic mood changes   |
| otm           | Very relaxed                  | Feels anxious                       |

Each of these was then converted into a “cost term” that could be fit into the Neuroticism “cost equation”. These are denoted as:

$$N_i = adp_i + (2 * wrd_i) + sdv_i + otm_i$$

Where for every  $i^{th}$  agent,  $adp_i$  = Adaptability Cost,  $wrd_i$  = Well-Rounded Cost,  $sdv_i$  = Steadiness Cost,  $otm_i$  = Optimism Cost

Through the cost equation, a feeling of urgency is being implemented during decision making. Higher value of  $N_i$  indicates poorer decision-making ability, which is then used to calculate an influence level.

This influence level can be later leveraged to calculate if the agent is able to influence others' decisions in a run-hide-fight scenario. The influence level is calculated as:

$$Influence = \begin{cases} \text{if } N_i > 0, |N_i + R_{lud}| \\ \text{else, } |N_i + R_{uud}| \end{cases}$$

Where  $R_{lud}$  is a random variable following a continuous uniform distribution over the lower set  $\{-0.5, \dots, 0.0\}$  and  $R_{uud}$  is a random variable following a continuous uniform distribution over the upper set  $\{0.0, \dots, 0.5\}$  both a part of the set  $\{-1.0, \dots, 1.0\}$  that denotes the continuous uniform distribution of the possible values of each cost term.

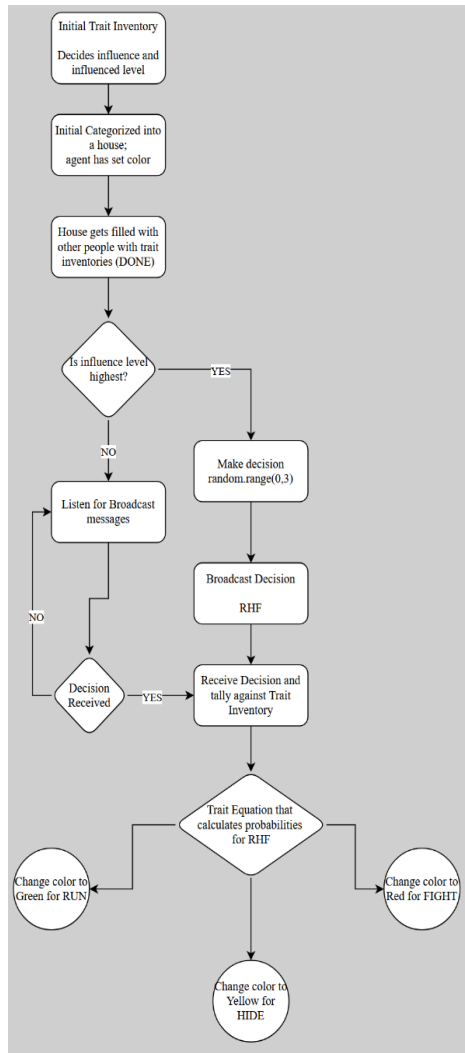


Figure 1: Simplified flowchart of simulation

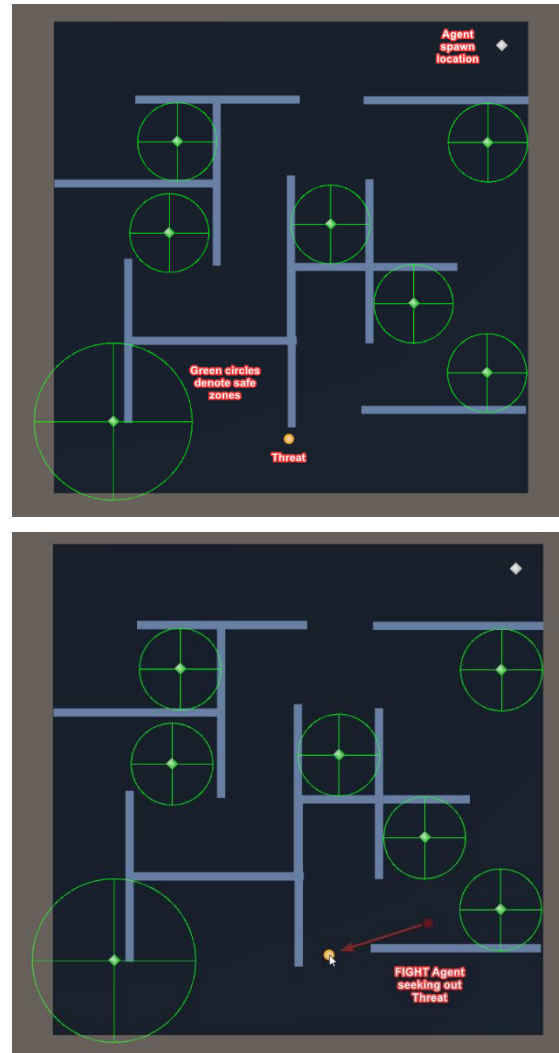


Figure 2: (Top) Setup in Unity engine; (Bottom) Agent chose FIGHT and seeking out Threat

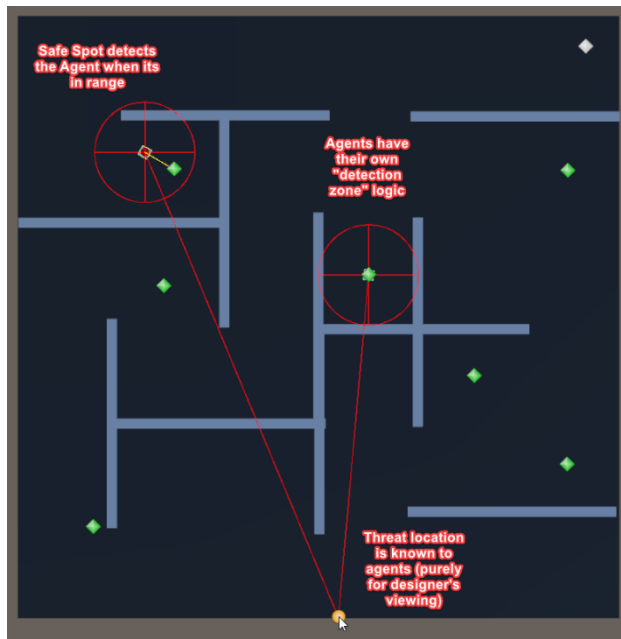


Figure 3: Setup during simulation run

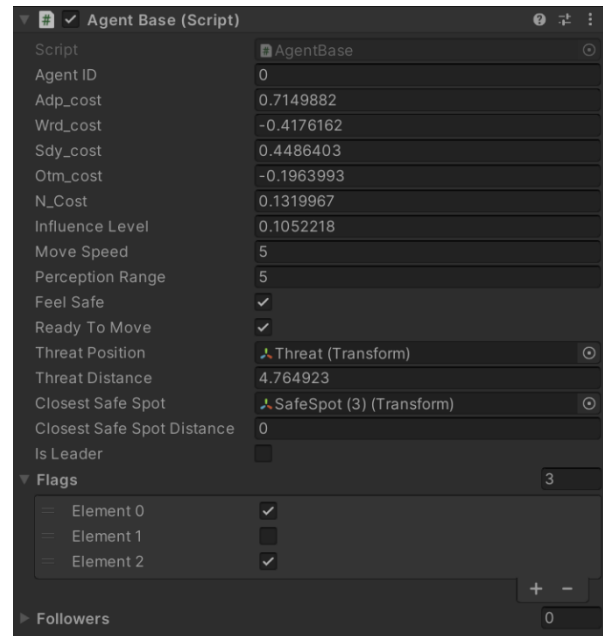


Figure 4: Agent script with Trait Inventory and variables needed for calculation during simulation

The influence score is appended with a randomized value, to introduce variation in the final stages of decision making and taking charge (Slowikowski, 2005). If the Neuroticism cost is high, the random variable can simulate “losing confidence” and if the Neuroticism cost is low, the random variable simulates “gaining confidence”, which introduces variability (a methodology of induced randomness in Monte Carlo Markov Chain optimization (Brooks, 1998).

### 3.2. Tools Used

The project was made using Unity version 2021.3.11f1, with the primary language of development as C#. For verification of agent-based methods and adoption logic AnyLogic Professional was used. The simulation was run on a laptop with 16 Gigabytes of Random Access Memory (RAM), an Intel i7 processor, and an RTX 2070 Graphics Processing Unit.

### 3.3. Methodology

Every agent in the simulation has a trait inventory. As discussed in Section 3.1. the direct result of the Trait Inventory scores is to establish an Influence Score, which is used to decide the “leader” among a group of agents.

The simulation first generates a set number of agents (configured via user input). The navigation agent component then assigns a random safe spot on the map. A “Safe Spot” is a place where the agents will evaluate their current state and decide from the available choices of RHF (Run, Hide, or Fight). The agent then moves to the initial spot, and a check is triggered where the agent calculates its Influence Level. Once that is done, the safe spot that has a list of all active agents in it will receive the Influence scores and designate a leader agent. Once a leader agent has been assigned, each of the members of the safe spot

will subscribe to the commands sent out by it, which represent one of the choices in RHF, and change their color according to three triage category colors (Melmer et al., 2019).

After the agent leader is assigned, the process is two-fold:

1. **Follower Agent:** “Subscribes” to the Leader Agent (using Unity’s Event System) and “Listens” to the responses issued by the leader. After initial assignment of safe spots, these agents calculate their own response to a perceived threat and store it in their “short-term memory”. Once the Leader Agent’s choice has been received, the follower agent – based on parameters such as Neuroticism score and Influence level – has the option to choose and follow the Leader Agent’s choice or choose for their own.
2. **Leader Agent:** Once the respective Safe Spot has assigned an agent a leader, based on weighted random function, they decide one of the three available actions. It then broadcasts the message to its follower agents (who are subscribers to it).

Once all agents are in a safe spot and a leader has been assigned, the simulation places the threat (or multiple threats) in a random position on the map. Based on the position, the safe spots change state from “Safe” to “Threat in Vicinity”. If the state of the safe spot changes to “Threat in Vicinity”, all agents within it execute their computed actions. The actions are defined as:

1. **RUN** – The agents change color to green and seek out the furthest available safe spot from the threat. Once the threat is in vicinity, they execute the action and move to the calculated safe spot.
2. **HIDE** – The agents change color to yellow and change their safe spot to another available safe spot on the map.
3. **FIGHT** – The agents change color to red and actively seek out the threat at their known position. Once they reach, a small event is run where the agent challenges against the threat’s threat level (pseudo-random number generation). Based on the outcome, the win/loss of the agent is recorded (win signifying they were able to overcome the threat and successfully subdue it).

Regardless of location of threat, all agents compute choices (i.e., change to a color to signify their decisions) and only if a safe zone is determined to have a “Threat in Vicinity”, the agents execute the action. After the completion of the action of the last agent, the simulation is notified and the threat is placed in a new location, and the agents compute their choices again. The evolution of the choices is recorded, and results are analyzed.

#### 4. Limitations and Assumptions

The personality model used by the agents in the simulation is not representative of a final model. Personality traits have influences on each other and could be quite complex to model. This proof-of-concept assumes that each personality trait described by the Big 5 Personality Model (McCrae & Costa, 1987; McCrae & John, 1992) exists independent of each other.

Another assumption for the proof of concept is that the system is closed and exists independent of external inputs i.e., there is no response force (police, firefighters, SWAT) to tackle the threat.



## 5. Future Work

The major future work for this proof of concept is to refine the digital personality model, to account for all possible traits and develop better cost equations to reflect decision making processes which are closer to real life.

Another major area that the project leaves scope for, is to try and add external factors to the model such as police response times, trained and untrained agents, building layouts, as well as availability of resources to the agents (resources can be defined as safety measures put in place, by the school or organization associated with the location).

Finally, this model only uses a part of the Big 5 Personality Model, whereas future iterations will include the other two models (Maslow's Hierarchy of Needs and DiSC model) as well.

## References

- Briggs, T. W., & Kennedy, W. G. (2016). In *2016 Winter Simulation Conference (WSC)* (pp. 3521–3531). Washington, DC; IEEE.
- Brooks, S. (1998). Markov chain Monte Carlo Method and its Application. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 47(1), 69–100. <https://doi.org/10.1111/1467-9884.00117>
- Byrne, K. A., Silasi-Mansat, C. D., & Worthy, D. A. (2015). Who chokes under pressure? the big five personality traits and decision-making under pressure. *Personality and Individual Differences*, 74, 22–28. <https://doi.org/10.1016/j.paid.2014.10.009>
- Cox, J. W., Rich, S., Chong, L., Trevor, L., Muyskens, J., & Ulmanu, M. (2018, April 20). More than 356,000 students have experienced gun violence at school since Columbine [web log]. Retrieved August 1, 2023, from <https://www.washingtonpost.com/education/interactive/school-shootings-database/>.
- Crooks, A. T., & Heppenstall, A. J. (2011). Introduction to agent-based modelling. *Agent-Based Models of Geographical Systems*, 85–105. [https://doi.org/10.1007/978-90-481-8927-4\\_5](https://doi.org/10.1007/978-90-481-8927-4_5)
- Denburg, N. L., Weller, J. A., Yamada, T. H., Shivapour, D. M., Kaup, A. R., LaLoggia, A., Cole, C. A., Tranel, D., & Bechara, A. (2009). Poor decision making among older adults is related to elevated levels of neuroticism. *Annals of Behavioral Medicine*, 37(2), 164–172. <https://doi.org/10.1007/s12160-009-9094-7>
- Digman, J. M. (1990). Personality structure: Emergence of the five-factor model. *Annual Review of Psychology*, 41(1), 417–440. <https://doi.org/10.1146/annurev.ps.41.020190.002221>
- Farmer, J., & Zhang, J. (2020). Run, Hide, and Fight to Save Your Life. *Florida Journal of Educational Research*, 58(7), 44–57. <https://doi.org/https://journals.flvc.org/fjer/article/download/133546/137465>

- Gun Violence Archive. (2023). *Mass Shootings in 2023 | Gun Violence Archive*. Gun Violence Archive. <https://www.gunviolencearchive.org/reports/mass-shooting?sort=desc&order=%23+Victims+Killed>
- History.com Editors (Ed.). (2009, November 9). *Columbine High School Shooting: Victims & Killers*. History.com. <https://www.history.com/topics/1990s/columbine-high-school-shootings>
- Hughes, H. P., Clegg, C. W., Robinson, M. A., & Crowder, R. M. (2012). Agent-based modelling and simulation: The potential contribution to organizational psychology. *Journal of Occupational and Organizational Psychology*, 85(3), 487–502. <https://doi.org/10.1111/j.2044-8325.2012.02053.x>
- Jackson, J. C., Rand, D., Lewis, K., Norton, M. I., & Gray, K. (2017). Agent-based modeling. *Social Psychological and Personality Science*, 8(4), 387–395. <https://doi.org/10.1177/1948550617691100>
- Jacobs, L. M., & Burns, K. J. (2017). The Hartford Consensus: Survey of the public and healthcare professionals on active shooter events in Hospitals. *Journal of the American College of Surgeons*, 225(3), 435–442. <https://doi.org/10.1016/j.jamcollsurg.2017.06.009>
- Kashima, Y. (2007). A social psychology of cultural dynamics: Examining how cultures are formed, maintained, and transformed. *Social and Personality Psychology Compass*, 2(1), 107–120. <https://doi.org/10.1111/j.1751-9004.2007.00063.x>
- Katsiyannis, A., Rapa, L. J., Whitford, D. K., & Scott, S. N. (2022). An examination of US school mass shootings, 2017–2022: Findings and implications. *Advances in Neurodevelopmental Disorders*, 7(1), 66–76. <https://doi.org/10.1007/s41252-022-00277-3>
- Klimstra, T. (2012). Adolescent personality development and Identity Formation. *Child Development Perspectives*, 7(2), 80–84. <https://doi.org/10.1111/cdep.12017>
- Lauriola, M., & Levin, I. P. (2001). Personality traits and risky decision-making in a controlled experimental task: An exploratory study. *Personality and Individual Differences*, 31(2), 215–226. [https://doi.org/10.1016/s0191-8869\(00\)00130-6](https://doi.org/10.1016/s0191-8869(00)00130-6)
- Lee, J. Y. (2019). *AGENT-BASED MODELING TO ASSESS THE EFFECTIVENESS OF RUN HIDE FIGHT* (thesis).
- Maslow, A. H. (1943). A theory of human motivation. *Psychological Review*, 50(4), 370–396. <https://doi.org/10.1037/h0054346>
- McCrae, R. R., & Costa, P. T. (1987). Validation of the five-factor model of personality across instruments and observers. *Journal of Personality and Social Psychology*, 52(1), 81–90. <https://doi.org/10.1037/0022-3514.52.1.81>
- McCrae, R. R., & John, O. P. (1992). An introduction to the five-factor model and its applications. *Journal of Personality*, 60(2), 175–215. <https://doi.org/10.1111/j.1467-6494.1992.tb00970.x>

- McLeod, S. (2007). Maslow's Hierarchy of Needs. *Simply Psychology*, 1, 1–18.  
[https://doi.org/https://www.simplypsychology.org/maslow.html?ez\\_vid=2cae626a2fe896279da43d587baa3eb663083817](https://doi.org/https://www.simplypsychology.org/maslow.html?ez_vid=2cae626a2fe896279da43d587baa3eb663083817)
- Melmer, P., Carlin, M., Castater, C. A., Koganti, D., Hurst, S. D., Tracy, B. M., Grant, A. A., Williams, K., Smith, R. N., Dente, C. J., & Sciarretta, J. D. (2019). Mass casualty shootings and emergency preparedness: A multidisciplinary approach for an unpredictable event. *Journal of Multidisciplinary Healthcare*, Volume 12, 1013–1021. <https://doi.org/10.2147/jmdh.s219021>
- Mendes, F. F., Mendes, E., & Salleh, N. (2019). The relationship between personality and decision-making: A systematic literature review. *Information and Software Technology*, 111, 50–71.  
<https://doi.org/10.1016/j.infsof.2019.03.010>
- Monden, Y. (1998). *Toyota production system: An integrated approach to just-in-time*. Engineering and Management Press.
- Nieto Fernández-Pacheco, A., Castro Delgado, R., Arcos González, P., Navarro Fernández, J. L., Cerón Madrigal, J. J., Juguera Rodríguez, L., Perez Alonso, N., Armero-Barranco, D., Lidon López Iborra, M., Damian, E. T., & Pardo Rios, M. (2018). Analysis of performance and stress caused by a simulation of a mass casualty incident. *Nurse Education Today*, 62, 52–57.  
<https://doi.org/10.1016/j.nedt.2017.12.016>
- Ozer, D. J., & Benet-Martínez, V. (2006). Personality and the prediction of consequential outcomes. *Annual Review of Psychology*, 57(1), 401–421.  
<https://doi.org/10.1146/annurev.psych.57.102904.190127>
- Pandya, A. (2014). *Bachelor of management studies unofficial portal*. BMS.  
<https://www.bms.co.in/simplest-way-to-help-you-learn-maslows-hierarchy-of-needs/>
- Price, L. A. (2015). (rep.). *DISC Instrument Validation Study* (pp. 1–16). People Keys.
- School Shooting Safety and Preparedness Act (2022). bill.
- Sloan, H. M. (2011). Responding to a multiple-casualty incident: Room For Improvement. *Journal of Emergency Nursing*, 37(5), 484–486. <https://doi.org/10.1016/j.jen.2010.07.013>
- Slowikowski, M. K. (2005). Using the disc behavioral instrument to guide leadership and Communication. *AORN Journal*, 82(5), 835–843. [https://doi.org/10.1016/s0001-2092\(06\)60276-7](https://doi.org/10.1016/s0001-2092(06)60276-7)
- Sugerman, J. (2009). Using the disc® model to improve communication effectiveness. *Industrial and Commercial Training*, 41(3), 151–154. <https://doi.org/10.1108/00197850910950952>
- U.S. Department of Homeland Security. (n.d.-a). *Active shooter - how to respond* - CISA. Active Shooter.  
<https://www.cisa.gov/sites/default/files/publications/active-shooter-how-to-respond-2017-508.pdf>

U.S. Department of Homeland Security. (n.d.-b). *Active shooter preparedness*. Active Shooter Preparedness | Cybersecurity and Infrastructure Security Agency CISA.  
<https://www.cisa.gov/topics/physical-security/active-shooter-preparedness>

U.S. Department of Homeland Security. (n.d.-c). *School and workplace violence*. School and Workplace Violence | Homeland Security. <https://www.dhs.gov/school-and-workplace-violence>